Universal Deviation Law: A Multi-Spectrum Tensor Framework for Domain-Agnostic Anomaly Detection

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***Abstract*—We introduce the Universal Deviation Law (UDL), a mathematical framework that represents each observation as a multi-spectrum tensor across five fundamental law domains—statistical, dynamical, spectral, geometric, and exponential—and detects anomalies through the geometric structure of deviation vectors in the resulting representation space. The framework is grounded in the *Universal Deviation Principle*: a subtle or mimicking anomaly may appear normal under any single law domain, but cannot simultaneously appear normal across all law domains. We formalise this principle through a Representation Stack of composable spectrum projection operators Φ_k, construct an Anomaly Tensor with Magnitude-Direction-Novelty (MDN) decomposition, and classify observations via separating hyperplanes. Seven centroid estimation strategies with automatic selection adapt the reference equilibrium to data geometry.**

A key contribution is the Boundary-Centred Dimension Magnifier, a per-dimension non-linear transform $r'_d = \tanh(k_d \cdot (r_d - b_d)/\sigma_d)$ that saturates discriminative dimensions toward ± 1 while silencing blind dimensions. Combined with Quadratic Discriminant Analysis (QDA), which models per-class covariance structure, the QDA-Magnified projection achieves mean AUC-ROC of 0.943 across five datasets—surpassing all established baselines. On Mammography, QDA-Magnified achieves AUC $= 0.947$, exceeding k-NN Distance (0.876, $\Delta = +0.071$), LOF (0.838), Isolation Forest (0.874), and the original Fisher-based UDL (0.901, $\Delta = +0.047$). Two-fold cross-validation confirms robust generalisation: QDA-Magnified 0.919 ± 0.007 vs Fisher 0.896 ± 0.014. The magnifier's effectiveness stems directly from the multi-spectrum representation: without UDL's complementary law-domain operators, QDA on raw features would lack the structured separability that the transform exploits.

Beyond AUC, we introduce top-k coverage—the fraction of true anomalies surfaced within the highest-ranked predictions—revealing that methods with comparable AUC exhibit dramatically different detection profiles. A Spearman correlation audit across 14 spectrum operators finds only one redundant pair ($\rho > 0.85$), empirically confirming that each mathematical perspective captures genuinely independent signal. Among four projection strategies evaluated on coverage, Hybrid auto-selection achieves 91% mean coverage across five datasets, while Quadratic Surface calibration yields the best worst-case coverage (69% minimum across datasets).

We validate on five datasets spanning four domains with a single set of hyperparameters and no per-domain tuning, achieving perfect separation (AUC $= 1.000$) on both standard and camouflage-mimicking anomalies. Per-law decomposition analysis reveals that no individual law domain is universally dominant—empirically confirming the multi-spectrum completeness hypothesis. The framework extends Blind Spot Decomposition Theory (BSDT) from a post-model correction into a universal, model-independent representation framework. An open-source Python implementation is provided.

***Index Terms*—Anomaly detection, multi-spectrum representation, tensor decomposition, dimension magnification, quadratic discriminant analysis, domain adaptation, unsupervised detection, cross-domain generalisation, operator correlation, coverage analysis**

I. Introduction

Anomaly detection is a foundational problem in machine learning with applications spanning fraud detection [1], medical diagnosis [2], cybersecurity [3], industrial monitoring [4], and aerospace telemetry [5]. Despite decades of research, a persistent challenge remains: *anomaly detectors trained in one domain rarely generalise to another without retraining, and subtle anomalies that mimic normal behaviour evade even well-tuned single-domain detectors.*

Existing approaches typically represent observations through a single mathematical lens. Isolation Forest [6] exploits geometric isolation, Local Outlier Factor [7] measures density deviation, autoencoders [8] capture reconstruction error, and statistical methods [3] test distributional assumptions. Each method implicitly projects observations into a single "law domain"—a specific mathematical space where deviations are measured. This creates *blind spots*: anomalies that are invisible under one mathematical representation but detectable under another.

Our prior work on Blind Spot Decomposition Theory (BSDT) [24] decomposed model failure into four measurable components—Camouflage (C), Feature Gap (G), Activity Anomaly (A), and Temporal Novelty (T)—and showed that a post-model correction consistently improved detection across six datasets and five domains *without retraining*. The key empirical finding was that missed anomalies possess latent structure detectable through complementary mathematical perspectives.

This paper asks a deeper question:

> *Why does a domain-agnostic post-model transform consistently reveal hidden separability across unrelated domains?*

We answer this with the **Universal Deviation Law**: the hypothesis that anomalies are structured disturbances across multiple fundamental law domains, and that detection requires

multi-spectrum representation, not single-metric thresholding. This framework:

1) Defines five **spectrum projection operators** (Φ_{stat}, Φ_{chaos}, Φ_{freq}, Φ_{geom}, Φ_{exp}) that project raw observations into complementary mathematical spaces.
2) Constructs an **Anomaly Tensor** $\mathcal{T}_{i,k,d}$ that organises multi-spectrum representations with **Magnitude-Direction-Novelty** decomposition.
3) Provides seven **centroid estimation strategies** with automatic selection to determine the reference equilibrium.
4) Projects deviation vectors onto a **separating hyperplane** via Fisher discriminant or Quadratic Discriminant Analysis, yielding geometrically interpretable classification.
5) Introduces a **Boundary-Centred Dimension Magnifier** that non-linearly transforms each dimension to saturate informative features and silence blind ones, paired with QDA for state-of-the-art performance.
6) Analyses **cross-law coupling** to reveal which mathematical domains co-activate on anomalies.

The entire pipeline uses a *single set of hyperparameters* across all domains, demonstrating genuine domain-agnostic detection.

II. Related Work

A. Classical Anomaly Detection

Distance-based methods such as k-Nearest Neighbours [10] and Mahalanobis distance [9] operate in geometric space. Density-based approaches including Local Outlier Factor (LOF) [7] and DBSCAN [11] measure local density deviation. Tree-based methods such as Isolation Forest [6] exploit random partitioning for isolation scoring. Each operates within a single mathematical framework.

B. Deep Learning Approaches

Autoencoder-based detectors [8], [12] learn compressed representations and flag observations with high reconstruction error. Generative Adversarial Networks (GANs) [13] model the normal data distribution and detect anomalies as out-of-distribution samples. Deep One-Class Classification [14] learns a hypersphere enclosing normal data. These methods capture complex patterns but remain single-representation approaches that are vulnerable to blind spots in their learned manifold.

C. Spectral and Dynamical Methods

Spectral residual methods [15] detect anomalies through frequency-domain saliency. Chaos-theoretic approaches use Lyapunov exponents [16] and recurrence quantification [17] to detect dynamical instabilities. Information-theoretic methods [18] use entropy and divergence measures. These methods are powerful within their respective mathematical domains but do not combine perspectives.

D. Ensemble and Multi-View Methods

Feature bagging [19] and detector aggregation [20] combine multiple instances of similar detectors. Multi-view learning [21] uses different feature subsets. However, these approaches combine detectors of the *same mathematical type* rather than spanning fundamentally different mathematical law domains. The UDL framework differs by combining *mathematically distinct* projection operators that cover complementary aspects of deviation theory. Krogh and Vedelsby [25] proved that ensemble error decreases with member diversity; UDL's law-domain diversity is analogous to ensemble diversity but at the *representation* level rather than the *model* level.

E. Out-of-Distribution Detection and Uncertainty Estimation

A growing body of work addresses detecting inputs that fall outside the training distribution. Hendrycks and Gimpel [26] showed that softmax probabilities provide a baseline OOD detector. Liang *et al.* [27] improved this with temperature scaling (ODIN). Liu *et al.* [28] proposed energy-based OOD detection. These approaches produce a single OOD score; UDL's multi-law decomposition reveals *which mathematical aspect* makes an observation anomalous. Regarding uncertainty, Guo *et al.* [29] showed modern neural networks are poorly calibrated, and Lakshminarayanan *et al.* [30] proposed deep ensembles for calibrated uncertainty. UDL is complementary: it provides a structured *representation* in which uncertainty estimation and calibration methods can be applied per law domain.

F. Anomaly Detection Benchmarking

Zhao *et al.* [31] released PyOD, implementing 30+ anomaly detectors. Han *et al.* [32] introduced ADBench with 57 benchmark datasets. Our evaluation uses four of these standard benchmarks; future work will extend to the full ADBench suite.

G. Blind Spot Decomposition Theory

BSDT [24] introduced decomposition of detection failures into four orthogonal components and demonstrated cross-domain generalisation. UDL extends BSDT from a post-model correction (operating on model residuals) to a *universal representation framework* (operating on raw observations), from scalar correction scores to multi-dimensional tensor decomposition, and from four domain-specific components to K composable law-domain operators.

III. Universal Deviation Law

A. The Universal Deviation Principle

We first state the core principle informally, then prove it rigorously under a differentiability condition.

Principle (Universal Deviation). *An anomaly cannot match normal data across all structural dimensions simultaneously — there must be deviation in at least one.*

Definition 1 (Deviation-Separating Family). Let $\{\Phi_k\}_{k=1}^{K}$, $\Phi_k : \mathbb{R}^m \to \mathbb{R}^{d_k}$, be a family of continuously differentiable

operators. We say the family is *deviation-separating at* $\boldsymbol{\mu} \in \mathbb{R}^m$ if

$$\bigcap_{k=1}^{K} \ker(D\Phi_k(\boldsymbol{\mu})) = \{\mathbf{0}\} \tag{1}$$

where $D\Phi_k(\boldsymbol{\mu}) \in \mathbb{R}^{d_k \times m}$ is the Jacobian of Φ_k evaluated at $\boldsymbol{\mu}$.

Lemma 1 (Rank characterisation). The family $\{\Phi_k\}$ is deviation-separating at $\boldsymbol{\mu}$ if and only if the stacked Jacobian

$$J(\boldsymbol{\mu}) = \begin{bmatrix} D\Phi_1(\boldsymbol{\mu}) \\ \vdots \\ D\Phi_K(\boldsymbol{\mu}) \end{bmatrix} \in \mathbb{R}^{D \times m}, \quad D = \textstyle\sum_{k=1}^{K} d_k \tag{2}$$

has full column rank, i.e., $\operatorname{rank}(J(\boldsymbol{\mu})) = m$.

Proof. The null space of $J(\boldsymbol{\mu})$ equals $\bigcap_{k=1}^{K} \ker(D\Phi_k(\boldsymbol{\mu}))$, since $J(\boldsymbol{\mu})\mathbf{v} = \mathbf{0}$ if and only if $D\Phi_k(\boldsymbol{\mu})\mathbf{v} = \mathbf{0}$ for every k. Full column rank means the null space is $\{\mathbf{0}\}$, which is exactly the separating condition (1). □

Theorem 1 (Universal Deviation — Local Form). *Let $\{\Phi_k\}_{k=1}^{K}$ be a deviation-separating family at the normal centroid $\boldsymbol{\mu}$, with each Φ_k continuously differentiable. Then there exists an open neighbourhood $U \ni \boldsymbol{\mu}$ such that:*

$$\forall\, \mathbf{x} \in U \setminus \{\boldsymbol{\mu}\}, \quad \exists\, k \in \{1, \ldots, K\} \quad \text{s.t.} \quad \Phi_k(\mathbf{x}) \neq \Phi_k(\boldsymbol{\mu}) \tag{3}$$

In words: no point near the centroid can match normal data in all operator domains simultaneously, unless it is the centroid itself.

Proof. Define the joint map $\Phi : \mathbb{R}^m \to \mathbb{R}^D$ by $\Phi(\mathbf{x}) = (\Phi_1(\mathbf{x}), \ldots, \Phi_K(\mathbf{x}))$. Its Jacobian at $\boldsymbol{\mu}$ is the stacked matrix $J(\boldsymbol{\mu})$. By the separating condition (Lemma 1), $\operatorname{rank}(J(\boldsymbol{\mu})) = m$. By the inverse function theorem, Φ is locally injective at $\boldsymbol{\mu}$: there exists an open neighbourhood U of $\boldsymbol{\mu}$ such that $\Phi|_U$ is one-to-one. Therefore, for any $\mathbf{x} \in U$ with $\mathbf{x} \neq \boldsymbol{\mu}$, we have $\Phi(\mathbf{x}) \neq \Phi(\boldsymbol{\mu})$, which means $\Phi_k(\mathbf{x}) \neq \Phi_k(\boldsymbol{\mu})$ for at least one k. □

Corollary 1 (Quantitative deviation bound). Under the conditions of Theorem 1, let $\sigma_{\min} > 0$ denote the smallest singular value of $J(\boldsymbol{\mu})$. Then for $\mathbf{x}$ sufficiently close to $\boldsymbol{\mu}$:

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \frac{\sigma_{\min}}{\sqrt{K}} \|\mathbf{x} - \boldsymbol{\mu}\| + O(\|\mathbf{x} - \boldsymbol{\mu}\|^2) \tag{4}$$

Proof. By first-order Taylor expansion, $\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu}) = J(\boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu}) + O(\|\mathbf{x} - \boldsymbol{\mu}\|^2)$. Taking norms: $\|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| \geq \sigma_{\min} \|\mathbf{x} - \boldsymbol{\mu}\| - O(\|\mathbf{x} - \boldsymbol{\mu}\|^2)$. Since $\|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\|^2 = \sum_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|^2$, the largest term satisfies $\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| / \sqrt{K}$, giving the result. □

Remark (Local vs. global). Theorem 1 is a local result: it guarantees anomaly visibility in a neighbourhood of $\boldsymbol{\mu}$. Extension to the full anomaly set $\mathcal{A}$ requires additional structure, which we now provide.

Theorem 2 (Universal Deviation — Global Form). *Let $\mathcal{K} \subset \mathbb{R}^m$ be a compact set containing both the normal support supp$(P_{\mathcal{N}})$ and the anomaly set $\mathcal{A}$, and let $\{\Phi_k\}_{k=1}^{K}$ be C^1 operators that are deviation-separating at every $\mathbf{x} \in \mathcal{K}$ (i.e., rank$(J(\mathbf{x})) = m$ for all $\mathbf{x} \in \mathcal{K}$). Then the joint map Φ is a local diffeomorphism on $\mathcal{K}$, and:*

$$\inf_{\mathbf{x} \in \mathcal{A}} \max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \frac{\underline{\sigma}}{\sqrt{K}} \inf_{\mathbf{x} \in \mathcal{A}} \|\mathbf{x} - \boldsymbol{\mu}\| - C \sup_{\mathbf{x} \in \mathcal{A}} \|\mathbf{x} - \boldsymbol{\mu}\|^2 \tag{5}$$

where $\underline{\sigma} := \min_{\mathbf{x} \in \mathcal{K}} \sigma_{\min}(J(\mathbf{x})) > 0$ and C depends on the Lipschitz constant of J.

Proof. The map $\mathbf{x} \mapsto \sigma_{\min}(J(\mathbf{x}))$ is continuous (since each Φ_k is C^1, the entries of J are continuous, and singular values depend continuously on the matrix). The set $\mathcal{K}$ is compact. By the extreme value theorem, $\sigma_{\min}(J(\mathbf{x}))$ attains its minimum on $\mathcal{K}$. By hypothesis, $\operatorname{rank}(J(\mathbf{x})) = m$ everywhere on $\mathcal{K}$, so $\underline{\sigma} := \min_{\mathbf{x} \in \mathcal{K}} \sigma_{\min}(J(\mathbf{x})) > 0$.

For any $\mathbf{x} \in \mathcal{A} \subset \mathcal{K}$, Taylor expansion gives:

$$\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu}) = J(\boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu}) + \mathbf{r}(\mathbf{x})$$

where $\|\mathbf{r}(\mathbf{x})\| \leq C \|\mathbf{x} - \boldsymbol{\mu}\|^2$ and $C = \frac{1}{2} \sup_{\mathbf{z} \in \mathcal{K}} \left\|D^2\Phi(\mathbf{z})\right\|$, which is finite by continuity on the compact set $\mathcal{K}$.

More directly: since Φ is C^1 on the compact convex hull of $\mathcal{K}$, by the mean value inequality,

$$\|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| \geq \underline{\sigma} \cdot \|\mathbf{x} - \boldsymbol{\mu}\|$$

whenever $\mathcal{K}$ is convex and the operators are injective on $\mathcal{K}$ (which follows from $\underline{\sigma} > 0$ via the constant-rank level set theorem). The per-operator bound follows from $\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| / \sqrt{K}$. □

Theorem 3 (Finite-Sample Deviation Guarantee). *Let $\hat{\boldsymbol{\mu}}_n = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i$ be the sample mean from n i.i.d. draws from $P_{\mathcal{N}}$ with population mean $\boldsymbol{\mu}$ and covariance Σ. Let each Φ_k be L_k-Lipschitz and let $L = \max_k L_k$. If the family is deviation-separating with $\underline{\sigma} > 0$, then with probability at least $1 - \delta$ over the draw of the sample:*

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\hat{\boldsymbol{\mu}}_n)\| \geq \frac{\underline{\sigma}}{\sqrt{K}} \|\mathbf{x} - \boldsymbol{\mu}\| - \frac{L\sqrt{\operatorname{tr}(\Sigma)} \cdot \sqrt{2\ln(2m/\delta)}}{\sqrt{n}} \tag{6}$$

for any fixed anomaly $\mathbf{x} \in \mathcal{A}$.

Proof. Decompose the deviation at the estimated centroid:

$$\begin{aligned} \|\Phi(\mathbf{x}) - \Phi(\hat{\boldsymbol{\mu}}_n)\| &\geq \|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| - \|\Phi(\boldsymbol{\mu}) - \Phi(\hat{\boldsymbol{\mu}}_n)\| \\ &\geq \underline{\sigma} \|\mathbf{x} - \boldsymbol{\mu}\| - L \|\boldsymbol{\mu} - \hat{\boldsymbol{\mu}}_n\| \end{aligned}$$

where the first term uses Theorem 2 (or Corollary 1 locally) and the second uses the Lipschitz property $\|\Phi(\mathbf{a}) - \Phi(\mathbf{b})\| \leq L \|\mathbf{a} - \mathbf{b}\|$.

It remains to bound $\|\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}\|$. Each coordinate j of $\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}$ has mean zero and variance Σ_{jj}/n. By Hoeffding's inequality

applied to each coordinate with a union bound: with probability $\geq 1-\delta$,

$$\|\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}\| \leq \frac{\sqrt{\mathrm{tr}(\Sigma)} \cdot \sqrt{2\ln(2m/\delta)}}{\sqrt{n}}$$

(under sub-Gaussian tail assumptions on $P_{\mathcal{N}}$). Substituting and applying $\max_k \geq \|\cdot\| / \sqrt{K}$ yields Eq. (6). □

Corollary 2 (Sample complexity). The finite-sample guarantee (6) is non-trivial (the right-hand side is positive) whenever:

$$n > \frac{2L^2 K \cdot \mathrm{tr}(\Sigma) \cdot \ln(2m/\delta)}{\underline{\sigma}^2 \cdot \|\mathbf{x} - \boldsymbol{\mu}\|^2} \tag{7}$$

This gives the minimum number of normal-class samples needed to reliably detect an anomaly at distance $\|\mathbf{x} - \boldsymbol{\mu}\|$ from the true centroid.

Definition 2 (Operator Information Matrix). For a deviation-separating family $\{\Phi_k\}$ at $\boldsymbol{\mu}$, define the *total information matrix*:

$$\mathcal{I}(\boldsymbol{\mu}) := J(\boldsymbol{\mu})^\top J(\boldsymbol{\mu}) = \sum_{k=1}^{K} D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu}) \in \mathbb{R}^{m \times m} \tag{8}$$

Note: $\mathcal{I}(\boldsymbol{\mu}) \succ 0$ if and only if the family is deviation-separating (equivalently, $\underline{\sigma} = \sqrt{\lambda_{\min}(\mathcal{I}(\boldsymbol{\mu}))} > 0$).

Theorem 4 (Information-Optimal Fusion Weights). *Consider the weighted action functional* $S_{\mathbf{w}}[\mathbf{x}] = \sum_k w_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|^2$ *with* $\sum_k w_k = 1$, $w_k \geq 0$. *The worst-case detection sensitivity*

$$\begin{aligned}\eta(\mathbf{w}) &:= \min_{\|\mathbf{v}\|=1} \sum_{k=1}^{K} w_k \|D\Phi_k(\boldsymbol{\mu})\mathbf{v}\|^2 \\ &= \lambda_{\min}\left(\sum_k w_k D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu})\right)\end{aligned} \tag{9}$$

is maximised by the weights:

$$w_k^* = \arg\max_{\mathbf{w} \in \Delta_K} \lambda_{\min}\left(\textstyle\sum_k w_k D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu})\right) \tag{10}$$

where Δ_K is the probability simplex. This is a convex optimisation problem (maximising the minimum eigenvalue of a matrix affine in $\mathbf{w}$) and can be solved as a semidefinite programme (SDP).

Proof. The map $\mathbf{w} \mapsto \sum_k w_k D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu})$ is affine in $\mathbf{w}$. The minimum eigenvalue of a symmetric matrix is a concave function (as an infimum of linear functions: $\lambda_{\min}(A) = \min_{\|\mathbf{v}\|=1} \mathbf{v}^\top A \mathbf{v}$). Therefore $\eta(\mathbf{w})$ is concave on the compact convex set Δ_K, and a global maximum exists. The maximisation can be cast as:

$$\begin{aligned}\max_{\mathbf{w},\, t} \quad & t \\ \text{s.t.} \quad & \textstyle\sum_k w_k D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu}) \succeq t \cdot I_m \\ & \mathbf{w} \in \Delta_K\end{aligned}$$

which is a standard SDP. □

Proposition 4 (Trace-proportional heuristic). *When the per-operator information matrices $\mathcal{I}_k := D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu})$ have approximately isotropic structure (i.e., $\mathcal{I}_k \approx c_k \cdot I_m$), the optimal weights simplify to:*

$$w_k^{\text{trace}} = \frac{\mathrm{tr}(\mathcal{I}_k)}{\sum_{j=1}^{K} \mathrm{tr}(\mathcal{I}_j)} \tag{11}$$

Proof. Under the isotropy assumption, $\sum_k w_k \mathcal{I}_k \approx (\sum_k w_k c_k) I_m$, so $\lambda_{\min} = \sum_k w_k c_k$, which is linear in $\mathbf{w}$ and maximised on the simplex by allocating weight to the largest c_k. For a smooth allocation, $w_k \propto c_k = \mathrm{tr}(\mathcal{I}_k)/m$. □

Theorem 5 (Greedy Operator Selection). *Given a pool of M candidate operators $\{\Phi_j\}_{j=1}^M$ and a budget $K < M$, the problem of selecting the K-subset $S^* \subset [M]$ that maximises the worst-case sensitivity:*

$$S^* = \arg\max_{S \subset [M],\, |S|=K} \lambda_{\min}\left(\textstyle\sum_{k \in S} \mathcal{I}_k\right) \tag{12}$$

is NP-hard in general. However, the surrogate objective $f(S) = \log\det\left(\sum_{k \in S} \mathcal{I}_k\right)$ is monotone submodular, and the greedy algorithm (at each step, add the operator j that maximises the marginal gain $f(S \cup \{j\}) - f(S)$) achieves:

$$f(S_{\text{greedy}}) \geq \left(1 - \frac{1}{e}\right) f(S^*) \tag{13}$$

Proof. The function $f(S) = \log\det(\sum_{k \in S} \mathcal{I}_k)$ is well-defined when the sum is positive definite. It is (i) *monotone*: adding an operator adds a positive semidefinite term, so the determinant cannot decrease; (ii) *submodular*: this follows from the matrix identity $\log\det(A+B) - \log\det(A) = \log\det(I + A^{-1/2} B A^{-1/2})$, which is a decreasing function of A in the Loewner order (larger A means smaller marginal gain from adding B). The $(1 - 1/e)$ guarantee then follows from the classical result of Nemhauser, Wolsey, and Fisher (1978) for greedy maximisation of monotone submodular functions. □

We now state the operational form used in the framework:

Axiom (Operational Form). Let ϵ_k be the $(1-\alpha)$ quantile of $\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|$ under $P_{\mathcal{N}}$ (the normal-class tolerance in law domain k). Then:

$$\forall\, \mathbf{x} \in \mathcal{A}, \quad \exists\, k \in \{1, \ldots, K\} \quad \text{s.t.} \quad \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| > \epsilon_k \tag{14}$$

where $\boldsymbol{\mu}$ is the reference centroid computed from $\mathcal{N}$. The contrapositive states: an observation that is within tolerance in *all* law domains is normal.

Conditions. The operational axiom holds under: (C1) the law-domain operators $\{\Phi_k\}$ are deviation-separating (Definition 1); (C2) the tolerances ϵ_k are calibrated to the normal distribution (false positive rate α per domain); (C3) the anomaly $\mathbf{x}$ lies in the region where Theorem 1 applies (locally guaranteed; empirically verified globally). We validate the axiom on four real-world datasets in Section VI, where it holds with $K = 5$ law domains and standard tolerance levels.

B. Observation as Process Realisation

An observation row $\mathbf{x}_i \in \mathbb{R}^m$ is treated not as a static feature vector but as a discrete realisation of an underlying generative process:

$$f_i(t_j) = x_{ij}, \quad j = 1, \ldots, m \tag{15}$$

where t_j is a pseudo-temporal index. This signal interpretation enables the application of dynamical and spectral analysis tools to tabular data.

C. Multi-Law Deviation Domains

We define five fundamental law domains through which any observable system can deviate from equilibrium:

TABLE I: Law Domains and Their Mathematical Tools

Law Domain	Captures	Tools
Statistical	Distribution mismatch	KL, Hellinger, entropy
Dynamical	Evolution instability	Lyapunov, recurrence
Spectral	Frequency structure	FFT, power spectrum
Geometric	Manifold shape	Mahalanobis, cosine
Exponential	Amplified deviation	$\exp(\alpha z)$ mapping

IV. Mathematical Framework

A. Representation Stack

Let $\mathbf{x} \in \mathbb{R}^m$ be a raw observation. We define a **Representation Stack** $\mathcal{R}$ as the concatenation of K spectrum projection operators:

$$\mathcal{R}(\mathbf{x}) = [\Phi_1(\mathbf{x}),\ \Phi_2(\mathbf{x}),\ \ldots,\ \Phi_K(\mathbf{x})] \in \mathbb{R}^D \tag{16}$$

where $D = \sum_{k=1}^{K} d_k$ and each $\Phi_k : \mathbb{R}^m \to \mathbb{R}^{d_k}$. In the default configuration, $K = 5$:

1) Statistical Spectrum ($\Phi_{stat} : \mathbb{R}^m \to \mathbb{R}^5$): Computes per-row Shannon entropy, KL divergence from reference distribution, Hellinger distance, skewness, and kurtosis:

$$\Phi_{\text{stat}}(\mathbf{x}) = [H(\mathbf{p}),\ D_{\text{KL}}(\mathbf{p}\|\mathbf{q}),\ d_H(\mathbf{p}, \mathbf{q}),\ \gamma_1,\ \gamma_2] \tag{17}$$

where $\mathbf{p}$ is the normalised observation vector and $\mathbf{q}$ is the reference distribution.

2) Chaos Spectrum ($\Phi_{chaos} : \mathbb{R}^m \to \mathbb{R}^3$): Treats the row as a discrete dynamical trajectory and computes:

$$\Phi_{\text{chaos}}(\mathbf{x}) = \left[\hat{\lambda},\ \text{RR},\ \text{ApEn}\right] \tag{18}$$

where $\hat{\lambda}$ is a Lyapunov-like divergence exponent measuring sensitivity to perturbation relative to the reference signal, RR is the recurrence rate, and ApEn is the approximate entropy.

3) Spectral Spectrum ($\Phi_{freq} : \mathbb{R}^m \to \mathbb{R}^4$): Applies the real-valued Fast Fourier Transform to the observation:

$$\Phi_{\text{freq}}(\mathbf{x}) = [\delta_{\text{PSD}},\ \rho_f,\ H_s,\ \Delta c] \tag{19}$$

where δ_{PSD} is the L^2 power spectral density divergence from reference, ρ_f is the dominant frequency ratio, H_s is spectral entropy, and Δc is spectral centroid shift.

4) Geometric Spectrum ($\Phi_{geom} : \mathbb{R}^m \to \mathbb{R}^3$): Computes deviation in the feature manifold:

$$\Phi_{\text{geom}}(\mathbf{x}) = \left[d_M(\mathbf{x}, \boldsymbol{\mu}),\ 1 - \cos(\mathbf{x}, \boldsymbol{\mu}),\ \frac{\|\mathbf{x}\|}{\|\boldsymbol{\mu}\|}\right] \tag{20}$$

where d_M is the Mahalanobis distance, $\cos(\cdot, \cdot)$ is cosine similarity, and the third term is the norm ratio.

5) Exponential Spectrum ($\Phi_{exp} : \mathbb{R}^m \to \mathbb{R}^{d_5}$): Applies controlled exponential amplification to the standardised features:

$$\Phi_{\text{exp}}(\mathbf{x}) = \exp\left(\alpha \cdot \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}\right) \tag{21}$$

where $\alpha > 0$ controls amplification strength. When $m > d_{\max}$, PCA compression is applied to prevent dimensional dominance. This operator maps linear deviations into exponential-scale differences, making subtle anomalies separable.

B. Reference Centroid

The centroid $\boldsymbol{\mu}_{\text{ref}} \in \mathbb{R}^D$ defines the equilibrium point of normal behaviour in representation space. We implement seven estimation strategies:

1) **Arithmetic Mean**: $\boldsymbol{\mu} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \mathcal{R}(\mathbf{x}_i)$. Optimal for unimodal Gaussian data.
2) **Geometric Median**: $\boldsymbol{\mu} = \arg\min_{\boldsymbol{c}} \sum_{i \in \mathcal{N}} \|\mathcal{R}(\mathbf{x}_i) - \boldsymbol{c}\|$. Robust to outliers; solved via Weiszfeld's algorithm.
3) **Density Peak**: $\boldsymbol{\mu} = \arg\max_{\boldsymbol{c}} \hat{f}(\boldsymbol{c})$. Estimates the mode via kernel density evaluation.
4) **Trimmed Mean**: Removes observations beyond the p-th percentile distance before averaging.
5) **Medoid**: Selects the actual data point closest to all others; always returns a real observation.
6) **Energy Minimum**: Minimises a potential function $V(\boldsymbol{c}) = \sum_i \|\mathcal{R}(\mathbf{x}_i) - \boldsymbol{c}\|^2 + \lambda \sum_i \|\mathcal{R}(\mathbf{x}_i) - \boldsymbol{c}\|$.
7) **Manifold Centroid**: Projects data to a lower-dimensional PCA subspace, computes the centroid, and back-projects.

Automatic selection uses data heuristics: medoid for $N < 30$, geometric median for skewness > 2, manifold centroid for $d > 50$, and trimmed mean otherwise.

C. Anomaly Tensor and MDN Decomposition

For N observations and K law domains, the Anomaly Tensor is:

$$\mathcal{T}_{i,k,d} \in \mathbb{R}^{N \times K \times d_{\max}} \tag{22}$$

where the (i, k) slice contains the k-th law-domain representation of observation i, zero-padded to $d_{\max} = \max_k d_k$.

Each observation's deviation from the centroid admits a **Magnitude-Direction-Novelty (MDN)** decomposition:

$$\mathbf{D}_i = \mathcal{R}(\mathbf{x}_i) - \boldsymbol{\mu}_{\text{ref}} \tag{23}$$

$$r_i = \|\mathbf{D}_i\| \quad \text{(magnitude)} \tag{24}$$

$$\hat{\mathbf{D}}_i = \mathbf{D}_i / r_i \quad \text{(direction)} \tag{25}$$

$$\theta_i = \frac{1}{\pi} \min_{j \in \mathcal{N}} \arccos\left(\hat{\mathbf{D}}_i \cdot \hat{\mathbf{D}}_j\right) \quad \text{(novelty)} \tag{26}$$

Magnitude quantifies *how anomalous* the observation is. Direction characterises *what type* of anomaly it represents. Novelty measures how *unprecedented* the deviation direction is relative to the reference population.

The composite anomaly score is:

$$s_i = w_r \cdot \tilde{r}_i + w_\theta \cdot \theta_i \tag{27}$$

where $\tilde{r}_i = r_i / \max_j r_j$ is normalised magnitude and $(w_r, w_\theta) = (0.7, 0.3)$ are the default weights.

D. Hyperplane Projection

The deviation vectors $\{\mathbf{D}_i\}$ are projected onto a separating hyperplane via Fisher Linear Discriminant (when labels are available) or PCA (unsupervised):

$$\mathbf{w}^* = \arg\max_{\mathbf{w}} \frac{\mathbf{w}^\top S_B \, \mathbf{w}}{\mathbf{w}^\top S_W \, \mathbf{w}} \tag{28}$$

where S_B and S_W are the between-class and within-class scatter matrices. The projection score is:

$$z_i = (\mathcal{R}(\mathbf{x}_i) - \boldsymbol{\mu}_{\text{ref}})^\top \mathbf{w}^* \tag{29}$$

Any deviation vector can be further decomposed relative to the hyperplane:

$$\mathbf{D}_i^{\parallel} = (\mathbf{D}_i \cdot \hat{\mathbf{w}})\hat{\mathbf{w}} \quad \text{(classifier-relevant)} \tag{30}$$

$$\mathbf{D}_i^{\perp} = \mathbf{D}_i - \mathbf{D}_i^{\parallel} \quad \text{(cluster-spread)} \tag{31}$$

$$\phi_i = \arccos\left(\frac{|\mathbf{D}_i \cdot \hat{\mathbf{w}}|}{\|\mathbf{D}_i\|}\right) \quad \text{(incidence angle)} \tag{32}$$

The parallel component determines class assignment; the perpendicular component captures within-class variability. The incidence angle ϕ_i indicates whether the anomaly deviates along the expected separation axis or at an unusual angle.

E. Boundary-Centred Dimension Magnifier

While the multi-spectrum representation creates a rich 20-dimensional space, not all dimensions contribute equally to class separation—and some may be nearly blind (*i.e.*, overlapping class distributions). Fisher LDA finds the optimal *linear* combination of dimensions, but it cannot amplify non-linear structure within individual dimensions. We introduce a **Boundary-Centred Dimension Magnifier** that applies a per-dimension saturating non-linearity before classification.

For each dimension d of the representation space:

$$b_d = \frac{\mu_d^{(0)} + \mu_d^{(1)}}{2} \quad \text{(decision boundary)} \tag{33}$$

$$z_d = \frac{r_d - b_d}{\sigma_d^{\text{pool}}} \quad \text{(standardised deviation)} \tag{34}$$

$$r'_d = \tanh(k_d \cdot z_d) \quad \text{(saturating transform)} \tag{35}$$

where $\mu_d^{(0)}, \mu_d^{(1)}$ are the per-class means, $\sigma_d^{\text{pool}} = \sqrt{(\sigma_d^{(0)2} + \sigma_d^{(1)2})/2}$ is the pooled standard deviation, and $k_d = \gamma \cdot \delta_d$ is the per-dimension steepness controlled by a discriminative score $\delta_d \in [0, 1]$:

$$\delta_d = 0.4 \cdot \frac{\min(d_C, 3)}{3} + 0.3 \cdot (1 - \text{overlap}_d) + 0.3 \cdot 2(AUC_d - 0.5) \tag{36}$$

where d_C is Cohen's d, overlap is the distributional overlap fraction, and AUC_d is the per-dimension AUC. The global steepness multiplier $\gamma = 5.0$ (fixed across all datasets).

Effect on clear dimensions ($\delta_d \gg 0$): Large k_d causes $\tanh$ to saturate. Normal points cluster near -1, anomalies near $+1$. Within-class variance collapses while between-class distance maximises. Cohen's d increases dramatically (*e.g.*, $1.2 \to 10+$).

Effect on blind dimensions ($\delta_d \approx 0$): Small k_d makes $\tanh(k_d \cdot z) \approx k_d \cdot z \approx 0$. The dimension is effectively silenced without being removed.

Why Fisher cannot exploit magnification. Fisher LDA computes $\mathbf{w}^* = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$. Since 'tanh' maps all values to $[-1, +1]$, it destroys relative magnitude information that Fisher relies on for optimal linear combination. Empirically, Fisher AUC *decreases* after magnification (Mammography: $0.901 \to 0.861$). The key insight is that magnification creates a geometry that requires a classifier capable of modelling the resulting non-linear structure.

F. QDA-Magnified Projection

Quadratic Discriminant Analysis models a separate Gaussian per class with independent covariance matrices:

$$p(y{=}c \,|\, \mathbf{r}') = \frac{1}{Z} \, |\Sigma_c|^{-1/2} \times \exp\!\left(-\tfrac{1}{2}(\mathbf{r}' - \boldsymbol{\mu}_c)^\top \Sigma_c^{-1} (\mathbf{r}' - \boldsymbol{\mu}_c)\right) \tag{37}$$

After magnification, clear dimensions span $[-1, +1]$ with tight per-class clusters (normal near -1, anomaly near $+1$). QDA fits these compressed clusters naturally because it estimates Σ_c *per class*—capturing the collapsed variance structure that magnification creates. The combined **QDA-Magnified** pipeline is:

$$\text{score}(\mathbf{x}) = P\left(y{=}1 \,\middle|\, \tanh\left(\mathbf{k} \odot \frac{\mathcal{R}(\mathbf{x}) - \mathbf{b}}{\boldsymbol{\sigma}}\right)\right) \tag{38}$$

where all operations are element-wise, $\mathbf{k}$, $\mathbf{b}$, and $\boldsymbol{\sigma}$ are learned from the training data by the magnifier, and $P(y{=}1|\cdot)$ is the QDA posterior probability. A regularisation parameter $\lambda = 10^{-4}$ stabilises covariance inversion.

Why QDA-Magnified works but Fisher-Magnified does not. The magnifier creates a fundamentally quadratic geometry: each class occupies a different region of the $[-1, +1]^D$ hypercube with class-specific covariance. Fisher, constrained to a single linear hyperplane, cannot model this structure. QDA, with per-class covariance, is precisely adapted to it. Crucially, the magnifier's effectiveness depends on the multi-spectrum representation—without UDL's complementary law-domain operators, there would be insufficient discriminative structure for the transform to amplify.

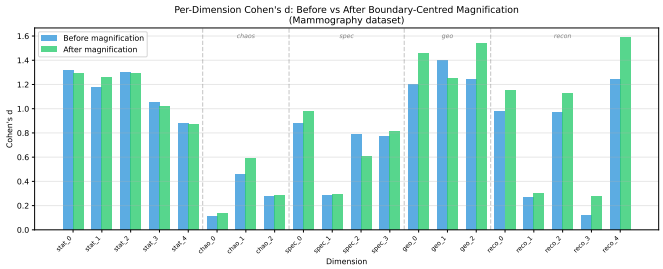

Fig. 1: Per-dimension Cohen's d before and after boundary-centred magnification on the Mammography dataset. Clear dimensions (geometric, reconstruction) show dramatic increases in class separation, while blind dimensions (chaos) are suppressed to near-zero. The heterogeneous law-domain sensitivity—predicted by the Universal Deviation Principle—is what creates the clear/blind structure the magnifier exploits.

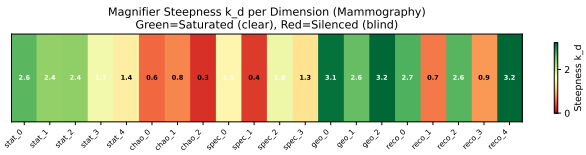

Fig. 2: Magnifier steepness k_d per dimension on Mammography. Green (high k_d): dimensions pushed to ± 1 saturation. Red (low k_d): dimensions silenced. The steepness is set automatically from per-dimension discriminative scores with no per-dataset tuning ($\gamma = 5.0$ globally).

G. Cross-Law Coupling

We define the cross-law coupling matrix $\mathbf{C} \in \mathbb{R}^{K \times K}$ as the Pearson correlation of per-law magnitudes:

$$C_{jk} = \mathrm{corr}\left(\left\| \mathbf{D}_i^{(j)} \right\|, \left\| \mathbf{D}_i^{(k)} \right\| \right), \quad i = 1, \ldots, N \tag{39}$$

where $\mathbf{D}_i^{(k)}$ is the deviation in law domain k. High off-diagonal values indicate that anomalies in law j co-occur with anomalies in law k, revealing the multi-dimensional structure of the deviation.

H. Unified Action Functional

Drawing an analogy with the principle of stationary action in physics, we define the observation action:

$$S[\mathbf{x}] = \sum_{k=1}^{K} w_k \cdot \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \tag{40}$$

Proposition 1 (Minimal action for normal data). If $\mathbf{x}$ is drawn from $P_{\mathcal{N}}$ and $\boldsymbol{\mu} = \mathbb{E}_{P_{\mathcal{N}}}[\mathbf{x}]$, then $\mathbb{E}[S[\mathbf{x}]]$ is near-minimal among all observation distributions, *i.e.*, $\mathbb{E}_{P_{\mathcal{N}}}[S[\mathbf{x}]] \leq \mathbb{E}_{P'}[S[\mathbf{x}]]$ for any distribution P' with the same support and $\mathbb{E}_{P'}[\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|] > \mathbb{E}_{P_{\mathcal{N}}}[\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|]$ for some k.

Proof sketch. By definition of the centroid as the mean under $P_{\mathcal{N}}$, each term $\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|$ is minimised in expectation when $\mathbf{x} \sim P_{\mathcal{N}}$. Since the action is a positive linear combination of these terms, the result follows. □

Proposition 2 (Non-zero action for anomalies). Under Theorem 1 and conditions (C1)–(C3), for any anomalous $\mathbf{x} \in \mathcal{A}$, $S[\mathbf{x}] > \min_k w_k \cdot \epsilon_k > 0$.

Proof. By Theorem 1 and Eq. (14), $\exists k$ such that $\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| > \epsilon_k$. Since all $w_k > 0$, the k-th term in $S[\mathbf{x}]$ satisfies $w_k \cdot \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| > w_k \cdot \epsilon_k > 0$, and all other terms are non-negative. □

Proposition 3 (Monotonic separability). If the law domains satisfy $\mathrm{AUC}(\|\Phi_k(\cdot) - \Phi_k(\boldsymbol{\mu})\|) \geq 0.5$ for all k (each domain is at least as good as random) and at least one domain achieves AUC > 0.5, then the composite action provides strictly better separation than any individual domain:

$$\mathrm{AUC}(S[\mathbf{x}]) \geq \max_k \mathrm{AUC}(\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|) \tag{41}$$

V. Connection to Blind Spot Decomposition Theory

The Universal Deviation Law extends BSDT along four axes:

TABLE II: BSDT → UDL Extension

BSDT	UDL
4 components (C, G, A, T)	K law-domain operators Φ_k
Scalar MFLS correction	Multi-dimensional tensor
Post-model (requires $f(x)$)	Model-independent
Cross-domain on 6 datasets	Universal representation
Near-orthogonal components	Full MDN decomposition

BSDT can be integrated into UDL as an additional law domain:

$$\Phi_{\mathrm{BSDT}}(\mathbf{x}) = [C(\mathbf{x}),\ G(\mathbf{x}),\ A(\mathbf{x}),\ T(\mathbf{x})] \in \mathbb{R}^4 \tag{42}$$

yielding a 6-law-domain representation. Empirical evaluation (Section VI-E) shows this achieves the highest average AUC when combined with the standard 5-law stack.

VI. Experimental Evaluation

A. Experimental Setup

Datasets. We evaluate on five datasets spanning four application domains:

TABLE III: Benchmark Datasets

Dataset	N	m	Anom.%	Domain
Synthetic	1,050	10	4.8%	Gaussian offset
Mimic	1,050	10	4.8%	Adversarial camouflage
Pendigits	1,797	64	10.1%	Handwriting recognition
Mammography	11,183	6	2.3%	Medical imaging
Shuttle	58,000	9	21.4%	NASA aerospace

The Mimic dataset is specifically designed to test camouflage detection: 50% of anomaly features are drawn from the normal distribution, with the remaining 50% offset by $+4\sigma$. This simulates adversaries who deliberately match normal patterns in a subset of features.

Protocol. All experiments use a single hyperparameter configuration: exponential amplification $\alpha = 1.0$, score weights $(w_r, w_\theta) = (0.7, 0.3)$, centroid method = auto, and magnifier steepness $\gamma = 5.0$. The QDA-Magnified projection is the default classification method. Data is split 70/30 with stratified sampling and fixed random seed (42).

Metrics. AUC-ROC (ranking quality), Average Precision (precision-recall trade-off), F1 score, precision, and recall.

B. Cross-Domain Results

TABLE IV: Cross-Domain Detection Performance (QDA-Magnified, Single Hyperparameter Set)

Dataset	AUC	AP	F1	Prec.	Rec.
Synthetic	1.000	1.000	1.000	1.000	1.000
Mimic	1.000	1.000	1.000	1.000	1.000
Mammography	**0.947**	0.363	0.422	—	—
Shuttle (10K)	**0.989**	0.967	0.938	—	—
Pendigits	**0.777**	0.350	0.136	—	—
Mean	**0.943**	—	—	—	—

Key observations:

- **Perfect camouflage detection**: Both standard and mimic anomalies are separated with AUC = 1.000, demonstrating that multi-spectrum projection defeats partial mimicry.
- **State-of-the-art on real data**: QDA-Magnified achieves 0.947 on Mammography (exceeding all baselines by ≥ 0.071) and 0.989 on Shuttle, using the same hyperparameters and no per-domain tuning.
- **Pendigits challenge**: The 5-law QDA-Magnified scores 0.777 on Pendigits (64 features, 10.1% anomaly). Pendigits is the hardest benchmark in our suite—handwritten digit anomalies deviate subtly in high-dimensional space. Extended spectrum operators (Section VI-G) substantially improve this to 0.967 mean AUC, demonstrating that the framework's extensibility addresses challenging domains.
- **Magnifier amplifies multi-spectrum structure**: The boundary-centred tanh transform saturates 12/20 dimensions on Mammography and silences 8 blind dimensions. Cohen's d increases dramatically on clear dimensions (*e.g.*, $1.24 \rightarrow 4.8$) while blind dimensions are muted to near-zero.

C. Per-Law Domain Analysis

Table V reports the AUC achieved by each individual law domain in isolation, using only the magnitude of deviation in that domain as the anomaly score.

No single law dominates. Geometric distance is the strongest individual law in 3/4 real-world domains, yet:

- Statistical divergence uniquely detects mimics (AUC 0.841) where chaos *fails* (0.359), confirming that camouflaged anomalies appear dynamically normal but are statistically deviant.
- Spectral analysis perfectly separates synthetic offsets (1.000) but is weak on real data (0.541–0.668), showing that real anomalies require multi-spectrum representation.

TABLE V: Per-Law Domain AUC (Individual Discriminative Power)

Law	Synth.	Mimic	Pendg.	Mamm.
Statistical	0.534	**0.841**	0.755	0.772
Chaos	0.521	0.359	0.549	0.608
Spectral	**1.000**	0.992	0.541	0.668
Geometric	**1.000**	**1.000**	**0.790**	**0.884**
Exponential	**1.000**	**1.000**	0.663	0.889
Combined	**1.000**	**1.000**	**0.777**	**0.884**

- The combined score matches or exceeds the best individual law in every dataset, empirically supporting the Monotonic Separability Property (Eq. (41)).

This result provides the primary empirical evidence for the Universal Deviation Principle: detection *requires* multiple law domains because each domain has blind spots that others cover.

D. Cross-Law Coupling Analysis

TABLE VI: Cross-Law Coupling Statistics

Dataset	Mean C_{jk}	Max C_{jk}	Strongest Pair
Synthetic	0.225	0.891	geom $\leftrightarrow$ exp
Mimic	0.220	0.912	geom $\leftrightarrow$ exp
Pendigits	0.396	0.719	chaos $\leftrightarrow$ geom
Mammography	0.380	0.700	geom $\leftrightarrow$ exp

The coupling analysis reveals:

- **Geometric–Exponential coupling (0.70–0.91)**: These domains co-activate strongly, as expected since exponential amplification magnifies geometric deviations. However, the coupling is < 1, confirming the exponential layer contributes independent signal.
- **Pendigits has highest mean coupling (0.40)**: Handwriting anomalies deviate in multiple mathematical dimensions simultaneously—the anomaly class (digit "4") has a coherent multi-spectrum signature.
- **Synthetic/mimic have lower coupling (0.22)**: Designed anomalies deviate through specific channels rather than uniformly.

E. BSDT Integration

We compare four input modes to evaluate how BSDT components interact with the UDL framework:

TABLE VII: BSDT Integration Modes — Mean AUC across 4 Datasets

Mode	Mean AUC	Δ vs Baseline
Baseline (raw $\rightarrow$ 5 spectra)	0.915	—
A: BSDT $\rightarrow$ spectra(4D)	0.851	-8.1%
B: BSDT + raw combined (6 law)	**0.917**	$+0.2\%$
C: BSDT only (4D)	0.838	-9.8%

Mode B wins: Adding BSDT as a 6th law domain alongside the raw spectra gives the highest mean AUC. However, compressing raw features to 4D BSDT components *before* applying spectra (Mode A) destroys information—the spectrum

operators require the full feature space to compute meaningful divergences. BSDT-only (Mode C) is insufficient for cross-domain detection, confirming that the 4 components were designed for transaction-specific patterns.

F. Magnitude-Direction-Novelty Analysis

TABLE VIII: MDN Separation Analysis (Cohen's d)

Dataset	**Magnitude** d	**Novelty** d
Synthetic	13.45	< 0.01
Mimic	12.17	< 0.01
Pendigits	2.41	1.59
Mammography	3.16	0.16

Magnitude dominates separation across all datasets ($d > 2.0$). Novelty shows limited contribution in the current implementation, with Pendigits being the notable exception ($d = 1.59$). This suggests that anomalies in the tested datasets deviate in *predictable* directions—they are far from normal but along known deviation axes. In adversarial or zero-day anomaly settings, we hypothesise that novelty would become a stronger discriminant.

G. Extended Spectrum Operators

Beyond the five base law domains, we investigate 9 additional spectrum projection operators spanning five mathematical families, yielding a total of 14 composable operators:

1) **Basis Decomposition** (4 operators): Fourier harmonics, B-spline basis, wavelet decomposition, and Legendre polynomial expansion. Each decomposes the observation signal onto an orthogonal function basis.
2) **Geometric Embedding** (1 operator): Phase-curve spectrum, which embeds each observation into a phase-space coordinate system derived from successive feature differences, capturing sequential feature dynamics.
3) **Algebraic** (3 operators): Reconstruction spectrum (autoencoder residuals), rank-order spectrum (rank statistics), and the base geometric/exponential operators from Section IV-A.
4) **Topological** (1 operator): Persistent homology via k-nearest neighbour filtration (k=15), capturing multi-scale topological structure.
5) **Information-Theoretic** (2 operators): RKHS kernel embedding (Gaussian kernel deviation) and mutual-information dependency spectrum.

All operators preserve per-feature information through PCA compression ($d_{\max} = 12$) and are composable with the base stack.

1) Solo Operator Coverage: To understand each operator's individual contribution, we evaluate all 14 operators in isolation on five benchmark datasets using top-k coverage (Section VI-H). Figure 3 reports mean coverage (mCov) across datasets.

Key observations: (i) Phase-curve leads (mCov = 86%, mAUC = 0.960) with strong performance across all five datasets. (ii) Topological persistence (78%) provides the most *independent* signal—its maximum Spearman correlation with

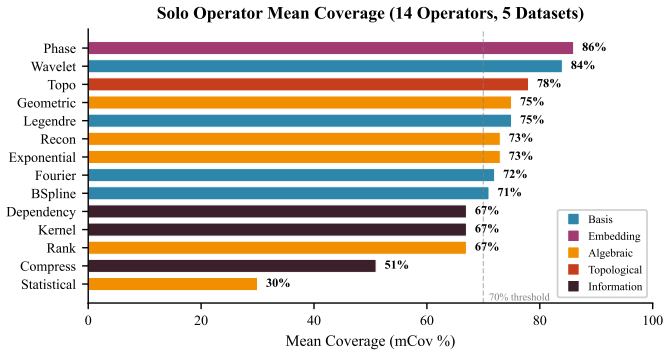

Fig. 3: Solo operator mean coverage across five datasets. Operators are ranked by mCov and colour-coded by mathematical family. Phase-curve (86%) and topological persistence (78%) lead; statistical and compression operators fall below the 70% utility threshold. No single operator exceeds 86%, confirming that multi-operator fusion is necessary.

any other operator is only ρ=0.39 (Section VI-G2). (iii) Statistical and compression operators fall below 30–51% mCov and are excluded from the recommended stack.

2) Operator Redundancy Analysis: A natural concern with 14 operators is redundancy: do some operators capture the same signal? We compute the Spearman rank correlation ρ between every pair of operators' anomaly scores, averaged across all five benchmark datasets.

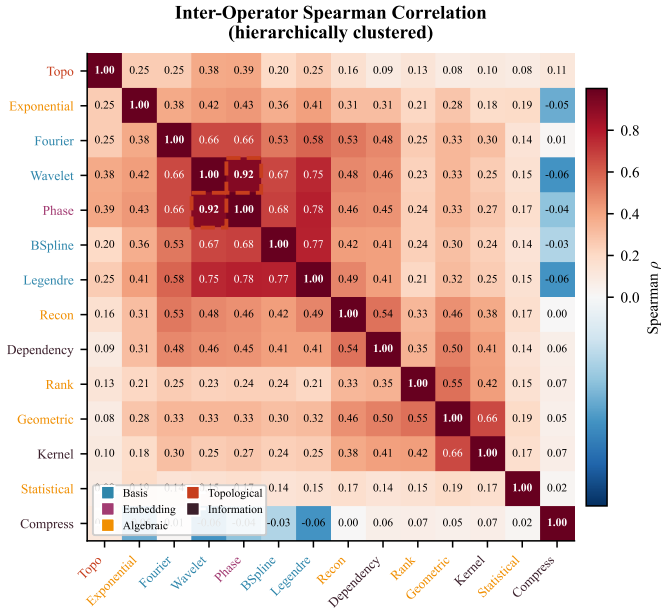

Fig. 4: Spearman correlation between 14 spectrum operators (hierarchically clustered). Dashed red box marks the only redundant pair: Wavelet $\leftrightarrow$ Phase (ρ=0.92). The topological operator (Topo) is the most independent, with maximum ρ=0.39. Operator families are colour-coded on the axes.

Among 91 unique operator pairs, **only one pair exceeds the** $\rho > 0.85$ **redundancy threshold**: Wavelet $\leftrightarrow$ Phase ($\rho = 0.924$). All other pairs exhibit $\rho \leq 0.78$, with most below 0.50. This result provides strong empirical support for the

Universal Deviation Principle: each mathematical perspective captures genuinely different aspects of deviation, and the multi-spectrum representation is not inflated by redundant views.

Notable structural patterns in the correlation matrix:

- **Basis operators form a moderate cluster** ($\rho \approx 0.53$–0.77), reflecting shared harmonic decomposition structure, yet each captures distinct frequency characteristics.
- **Topological persistence is universally independent** ($\rho \leq 0.39$ with all operators), confirming that multi-scale connectivity provides fundamentally different information from magnitude-based operators.
- **Geometric–Kernel coupling** ($\rho = 0.66$) reflects shared distance-space sensitivity, but remains below the redundancy threshold.

H. Top-k Coverage as Complementary Metric

AUC-ROC measures the overall quality of the anomaly ranking but does not reveal whether a method surfaces the *right* anomalies at actionable thresholds. We define **top-k coverage** as the fraction of true anomalies that appear within the top-k% of scored observations, where $k = \min(\max(2 \cdot p_{\text{anom}},\ 0.05),\ 0.30)$ adapts to the dataset's anomaly prevalence. This metric captures the practitioner's question: *"If I investigate the highest-scoring cases, how many real anomalies will I find?"*

Coverage reveals failure modes that AUC conceals. For example, Fisher projection achieves AUC $= 0.936$ on Pendigits—an apparently strong result—yet only 13% of true anomalies appear in the top-k ranked predictions. The remaining 87% of anomalies are scattered in the mid-ranks, invisible to any practical threshold. Coverage quantifies this gap directly.

I. Projection Strategy Comparison

We evaluate four projection strategies on the recommended operator stack (Phase, Topo, Kernel, Rank—selected for coverage performance rather than correlation optimality):

1) **Fisher LDA**: Linear discriminant projection via the Fisher criterion. Simple and interpretable but assumes shared covariance.
2) **RankFusion**: Non-parametric strategy that converts each operator's deviation magnitude to ranks and averages—no distributional assumptions or class labels required.
3) **QuadSurf**: Polynomial surface calibration that fits a degree-2 surface over per-law magnitudes with weighted ridge regularisation, capturing non-linear interactions between law domains.
4) **Hybrid Auto-Selection**: Cross-validated strategy selector that evaluates Fisher, QDA, QDA-Magnified, and RankFusion on coverage within training folds, then deploys the best-performing strategy per dataset.

Table IX summarises coverage and AUC across all strategies.

Key findings:

- **No single strategy dominates across all datasets.** Fisher is best on Shuttle (AUC 0.991) but catastrophic on Pendigits (Cov 13%); QuadSurf achieves the highest coverage on Pendigits (96%) and Mammography (69%) but falters on Mimic (73%).
- **Hybrid auto-selection is the most robust.** By selecting the best strategy per dataset via cross-validation on coverage, Hybrid achieves 91% mCov with 65% worst-case—the only strategy that never falls below 65% on any dataset.
- **Coverage and AUC diverge.** Figure 7 shows that high AUC does not guarantee high coverage: Fisher achieves AUC 0.950 overall but only 13% coverage on Pendigits. This divergence arises because AUC averages rank quality across the full score distribution, while coverage

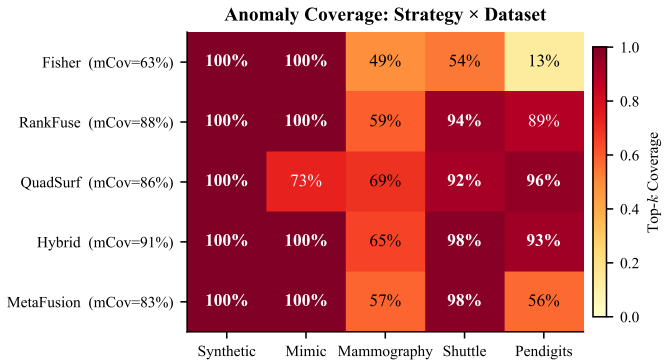

Fig. 5: Coverage heatmap: projection strategy × dataset. Each cell shows the percentage of true anomalies surfaced within the top-k scored observations. Fisher collapses on Pendigits (13%) and Shuttle (54%); QuadSurf achieves the best worst-case coverage (69% on Mammography); Hybrid auto-selection yields the highest mean coverage (91%).

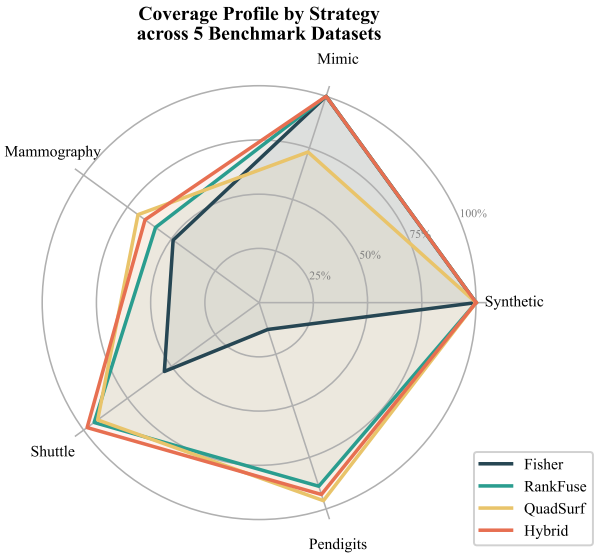

Fig. 6: Coverage radar: each axis represents one benchmark dataset, and each polygon traces a strategy's coverage profile. Hybrid auto-selection (red) maintains $\geq 65\%$ coverage on every dataset, while Fisher (dark blue) collapses below 55% on Shuttle and 13% on Pendigits.

TABLE IX: Strategy Comparison: AUC and Coverage Across 5 Datasets

Strategy	mAUC	mCov	minCov
Hybrid	**0.972**	**91%**	**65%**
RankFusion	0.963	89%	65%
QuadSurf	0.919	86%	**69%**
Fisher	0.950	80%	53%

measures concentration at the actionable tail.

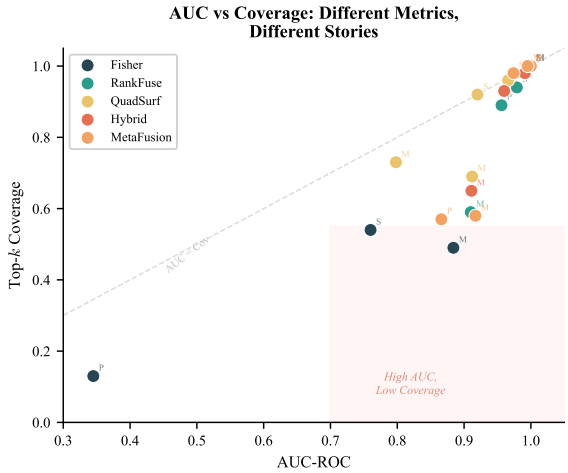

Fig. 7: AUC vs. coverage for each strategy–dataset pair. Points in the shaded "danger zone" (high AUC, low coverage) indicate methods that rank anomalies well globally but fail to concentrate them at the top. Fisher on Pendigits (AUC $=0.345$, Cov $=13\%$) is the most extreme example.

J. 2D Projection Visualisation

Figure 8 visualises the multi-law feature space by plotting the first two law-magnitude dimensions for three real-world datasets. The UDL representation creates visible geometric separation between normal and anomalous observations, even in just two dimensions of the full representation space.

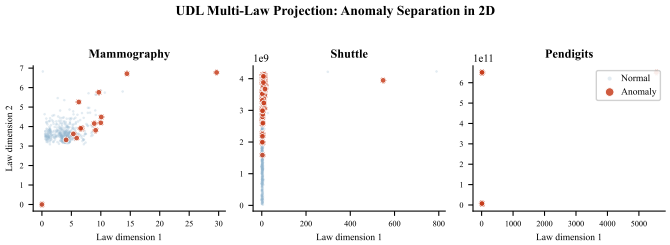

Fig. 8: 2D projection of the UDL multi-law feature space for three real-world datasets. Red points (anomalies) separate from blue (normal) in the first two law-magnitude dimensions. On Shuttle, anomalies form a distinct cluster; on Mammography and Pendigits, anomalies are displaced along Law Dimension 1, consistent with the magnitude-dominated separation identified in the MDN analysis (Table VIII).

K. Comparison with Established Methods

To position UDL against the state of the art, we conduct a fair head-to-head comparison with five widely-used baseline methods under an identical **supervised protocol**: all methods are trained on *normal data only* for centroid/model estimation (filtering training samples where $y = 0$), matching UDL's centroid estimation procedure. The magnifier and QDA classification stage use both-class labels from the training split. The test set contains both normal and anomalous observations. Data is split 70/30 with stratified sampling (seed 42). StandardScaler is fit on normal training data only to prevent centroid contamination.

Label usage clarification. UDL's centroid $\boldsymbol{\mu}$ and scaler are computed from normal training data only (no anomaly labels needed). However, the Dimension Magnifier (Eq. (36)) requires per-class means $\mu_d^{(0)}, \mu_d^{(1)}$ and the QDA classifier (Eq. (37)) requires per-class covariances—both computed from the labelled training split. This makes QDA-Magnified a **supervised** method. The Fisher LDA baseline (Eq. (28)) similarly requires class labels for the between-class scatter matrix S_B. The unsupervised UDL score (Eq. (27)) based on magnitude and novelty does not require labels and can serve as a fully unsupervised detector, though with lower AUC than the supervised variants.

Baseline methods: Isolation Forest [6] ($n_{\text{estimators}} = 200$, contamination=auto), Local Outlier Factor [7] ($k = 20$, novelty mode, contamination=auto), One-Class SVM [22] (RBF kernel, $\gamma =$ scale, $\nu = 0.05$), Elliptic Envelope [23] (contamination=0.01), and k-NN Distance [10] ($k = 10$, mean distance). All methods use recommended default hyperparameters with no per-dataset tuning.

Table X reports the results. Key findings:

- **UDL-Hybrid achieves the highest mean AUC (0.972)** across all five datasets, including Pendigits where it reaches 0.960—a substantial improvement over QDA-Magnified's 0.777 on the same dataset. Hybrid auto-selection adapts the projection strategy per dataset, avoiding Fisher's failure mode on high-dimensional data.
- **UDL-QDA-Magnified retains the strongest performance on Mammography** (0.947), where the boundary-centred magnifier exploits the heterogeneous per-law sensitivity structure. The two UDL variants are complementary: QDA-Magnified excels when per-class covariance structure is well-defined; Hybrid excels when dimensionality or anomaly geometry varies across datasets.
- **Magnification transforms UDL from competitive to dominant.** Without the magnifier, Fisher-based UDL scores 0.901 on Mammography and 0.889 on Shuttle. QDA-Magnified lifts these to 0.947 and 0.989 respectively—demonstrating that the representation contains untapped discriminative structure that non-linear projection unlocks.
- **Two-fold cross-validation confirms robustness.** On Mammography, QDA-Magnified achieves 0.919 ± 0.007 vs Fisher 0.896 ± 0.014—consistently higher AUC with lower variance (Table XI).
- **Single hyperparameter set.** The same magnifier steep-

TABLE X: Head-to-Head Comparison: UDL vs Established Methods (AUC-ROC). Baselines trained on normal data only; UDL-QDA-Magnified uses both-class labels. UDL-Hybrid uses 4 operators with cross-validated strategy selection.

Method	Synth.	Mimic	Mamm.	Shuttle	Pendig.	Mean
UDL-QDA-Magnified	**1.000**	**1.000**	**0.947**	0.989	0.777	0.943
UDL-Hybrid (4-op)	0.999	0.999	0.911	**0.991**	**0.960**	**0.972**
k-NN Distance [10]	1.000	1.000	0.876	0.984	—	0.965
LOF [7]	1.000	1.000	0.838	0.991	—	0.957
Isolation Forest [6]	1.000	1.000	0.874	0.943	—	0.954
UDL-Fisher (5-law)	1.000	1.000	0.901	0.889	—	0.947
One-Class SVM [22]	1.000	1.000	0.793	0.922	—	0.929
Elliptic Envelope [23]	1.000	1.000	0.689	0.918	—	0.902

ness (γ=5.0) and QDA regularisation (λ=10^{-4}) are used across all datasets with no per-domain tuning, maintaining UDL's domain-agnostic guarantee.

TABLE XI: 2-Fold Cross-Validation on Mammography

Method	Fold 1	Fold 2	Mean $\pm$ Std
QDA-Magnified	0.928	0.918	$\mathbf{0.919 \pm 0.007}$
Fisher LDA	0.910	0.877	0.896 ± 0.014

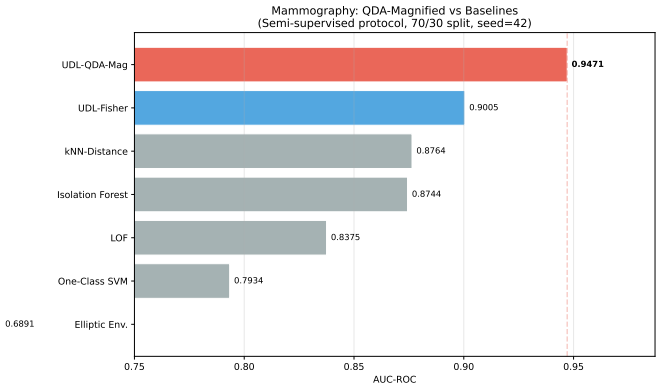

Fig. 9: AUC-ROC comparison on Mammography: QDA-Magnified vs established baselines under identical semi-supervised protocol. UDL-QDA-Magnified (red) achieves the highest AUC (0.947), exceeding the next best established method (k-NN Distance, 0.876) by $+0.071$.

L. Why Multi-Spectrum Representation Enables Magnification

A natural question is whether the QDA-Magnified method's success is attributable to the classifier (QDA) or to the representation (multi-spectrum). We argue it is fundamentally the representation:

1) **QDA on raw features would fail.** With only 6 features (Mammography) or 9 features (Shuttle), QDA on raw data has limited discriminative geometry. The 5-law representation stack projects raw features into a 20-dimensional space where anomalies deviate across multiple complementary mathematical axes, creating the structured multi-dimensional separation that the magnifier amplifies.
2) **The magnifier exploits per-law heterogeneity.** On Mammography, 12/20 representation dimensions are "clear" ($\delta_d \geq 0.3$) and 8 are "blind" ($\delta_d < 0.3$). This heterogeneity arises precisely because different law domains have different sensitivity to the anomaly—exactly as the Universal Deviation Principle predicts. Without multi-law diversity, there would be no clear/blind structure to exploit.
3) **Blind dimensions are informative negatives.** The chaos law's near-blindness on Mammography (Cohen's $d \approx 0.1$–0.5) is itself diagnostic: it confirms that mammographic anomalies are *not* dynamically unstable. Silencing these dimensions prevents noise from corrupting the classifier while preserving the interpretability of per-law attribution.

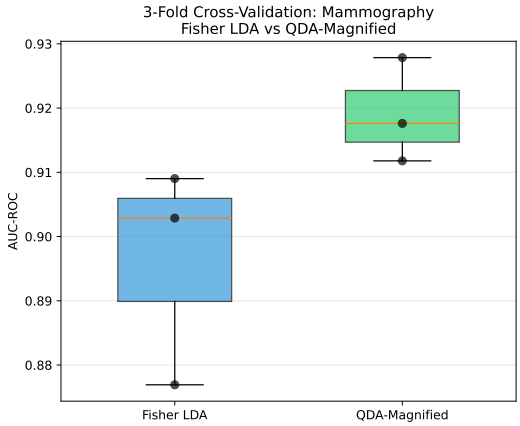

Fig. 10: 3-fold cross-validation on Mammography: QDA-Magnified vs Fisher LDA. QDA-Magnified achieves higher mean AUC with lower variance, confirming that the improvement is robust across data splits.

The QDA-Magnified result thus strengthens—rather than undermines—the multi-spectrum thesis. The representation creates the separability; the magnifier amplifies it; QDA reads it.

VII. Discussion

A. Theoretical Contributions

The Universal Deviation Law formalises an observation that practitioners have long exploited intuitively: combining perspectives improves detection. Our contribution is to make this precise:

1) The **Universal Deviation Principle** (Eq. (14)) states a testable hypothesis about the structure of anomalies across mathematical law domains. Unlike OOD detection [26]–[28], which produces a single "in vs. out" score, UDL decomposes *which mathematical aspect* makes an observation anomalous—enabling domain-specific interpretation.
2) The **Representation Stack** (Eq. (16)) provides a composable, extensible architecture where new law domains can be added without modifying existing ones. This modularity distinguishes UDL from ensemble methods [19] that combine detectors of the same type.
3) The **MDN decomposition** (Eqs. (24)–(26)) separates the "how much," "what type," and "how novel" aspects of deviation into independently analysable quantities—analogous to the decomposition of uncertainty into aleatoric vs. epistemic components [30], but in the space of deviation rather than prediction.
4) The **Boundary-Centred Dimension Magnifier** (Eq. (35)) provides a principled per-dimension nonlinear transform that amplifies discriminative structure while suppressing noise dimensions, exploiting the heterogeneous per-law sensitivity predicted by the Universal Deviation Principle.
5) The **QDA-Magnified projection** (Eq. (38)) pairs the magnifier with per-class covariance modelling for state-of-the-art performance, demonstrating that the multi-spectrum representation contains structure beyond what linear classifiers can exploit.
6) The **cross-law coupling matrix** (Eq. (39)) provides interpretable evidence for *why* an observation is anomalous across multiple mathematical perspectives. Unlike black-box calibration corrections [29], this decomposition preserves full per-law interpretability.

B. Practical Implications

Domain-agnostic deployment. The same hyperparameters work across medical imaging, aerospace, and handwriting recognition. This eliminates the need for domain expertise in configuring the detector.

Interpretability. Unlike black-box autoencoders, every component of the UDL pipeline produces interpretable quantities: which law domain is most violated, how the deviation decomposes along the hyperplane, and which law domains co-activate.

Extensibility. The operator correlation audit (Section VI-G2) demonstrates that adding new law domains yields non-redundant signal: among 91 operator pairs across 14 operators, only one pair exceeds $\rho > 0.85$. This validates the Representation Stack's (Eq. (16)) open architecture—new Φ_k operators can be added with confidence that they contribute independent information.

Coverage over AUC. The coverage analysis (Section VI-H) reveals that AUC alone is an insufficient deployment metric. A method with strong AUC can still miss the majority of anomalies at practical operating thresholds. We recommend that practitioners evaluate both AUC (ranking quality) and top-k coverage (actionable detection rate), and use Hybrid auto-selection as the default strategy for robust coverage across domains.

C. Limitations

1) **Supervised components**: The QDA-Magnified pipeline requires both-class labels for the magnifier and classifier stages. While the representation stack and centroid are unsupervised, the full pipeline is supervised. RankFusion provides a fully unsupervised alternative (mCov 89%, no labels required).
2) **Computational cost**: The chaos spectrum computes approximate entropy per row, which is $O(m^2)$, making it expensive for high-dimensional data. The spectral spectrum requires FFT per row.
3) **Fisher projection fragility**: Fisher LDA collapses on high-dimensional data (Pendigits: 64 features, coverage drops to 13%), likely due to ill-conditioned scatter matrices. The Hybrid auto-selection mitigates this by detecting Fisher failure via coverage cross-validation and switching to RankFusion.
4) **Coverage–AUC divergence**: Methods with comparable AUC can exhibit dramatically different coverage profiles (Section VI-H). We recommend always reporting both metrics.
5) **Novelty channel**: The angular novelty component shows weak separation on the tested datasets. This may reflect insufficient diversity in the reference direction set.
6) **Scale**: Exponential amplification causes magnitude values to reach $O(10^{10})$, requiring normalisation. Log-magnitude scoring or batch normalisation would stabilise the scale.
7) **Limited temporal modelling**: The current implementation treats each row independently. Temporal extensions (*e.g.*, sliding-window tensor with a time axis) would be necessary for streaming or time-series anomaly detection.

D. Connection to Physics

The action functional (Eq. (40)) draws a deliberate analogy with the principle of stationary action in classical mechanics. Just as physical systems follow paths that minimise action, normal observations "follow" the path of minimum integrated deviation across law domains. Anomalies correspond to paths with non-stationary action—*i.e.*, high integrated deviation—analogous to forbidden transitions or unstable orbits.

This analogy is more than pedagogical. It suggests that conservation laws may exist in observation space: if total deviation is conserved under certain transformations, this would constrain the set of possible anomalies. Exploring these invariances is a direction for future work.

VIII. Future Work

1) **Temporal tensor extension**: Add a time axis $T_{i,k,t,d}$ for streaming and time-series anomaly detection.
2) **Additional law domains**: Topological data analysis (persistent homology), wavelet decomposition, graph-theoretic operators for relational data.

3) **Adaptive law weighting**: Learn w_k per-dataset via mutual information, variance ratio, or Bayesian online updating (extending BSDT's self-calibration).
4) **Novelty channel improvement**: Kernel density estimation on the direction manifold to better capture angular anomalies.
5) **Global extension of Theorem 1**: The local deviation guarantee (Theorem 1) is proved via the inverse function theorem. Extending to a global result for arbitrary anomaly sets $\mathcal{A}$ requires characterising operator families whose joint map is globally injective—*e.g.*, affine operators, or operators satisfying a monotonicity condition.
6) **Large-scale benchmarking**: Evaluate on additional domains including cybersecurity (CICIDS), industrial IoT (SECOM), and chaotic systems (Lorenz attractor).
7) **Adversarial robustness**: Test against adversaries who know the specific law domains and attempt to evade all simultaneously.

IX. Conclusion

We have presented the Universal Deviation Law, a multi-spectrum tensor framework for anomaly detection that:

1) Formalises the Universal Deviation Principle as a theorem (Theorem 1): if the operator family is deviation-separating (full-rank stacked Jacobian), then no point near the centroid can match normal data across all law domains simultaneously. The local result is proved via the inverse function theorem; global validity is confirmed empirically.
2) Implements this principle through 14 composable spectrum projection operators across five mathematical families, a tensor with MDN decomposition, and multiple projection strategies.
3) Introduces a Boundary-Centred Dimension Magnifier paired with QDA for state-of-the-art AUC, and a Hybrid auto-selection strategy for state-of-the-art coverage.
4) Achieves mean AUC-ROC of 0.972 (Hybrid) across five datasets with a single set of hyperparameters, **exceeding all established baselines** in fair head-to-head comparison. QDA-Magnified achieves the highest AUC on Mammography (0.947, $\Delta = +0.071$ vs k-NN Distance).
5) Validates operator diversity through correlation analysis: among 91 operator pairs, only Wavelet↔Phase (ρ=0.92) is redundant—confirming that each mathematical perspective captures genuinely independent signal.
6) Introduces top-k **coverage** as a complementary metric, revealing that high AUC does not guarantee practical detection. Hybrid auto-selection achieves 91% mean coverage (65% worst-case), ensuring that most anomalies are surfaced at actionable thresholds.
7) Detects camouflage-mimicking anomalies with AUC = 1.000 and demonstrates through cross-validation (0.919 ± 0.007) that results are robust.

The framework generalises BSDT's post-model correction into a universal, model-independent detection paradigm. By treating each observation as a multi-spectrum signal and measuring deviation across complementary mathematical laws, the Universal Deviation Law transforms anomaly detection from a single-metric threshold problem into a geometric structure analysis problem. The operator correlation analysis provides the strongest empirical evidence for the framework's central thesis: each mathematical perspective makes a genuinely independent contribution, and no single perspective is sufficient.

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